Muhammad Raihan Thaffan Hidayat

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Profile

Telecommunications Engineering graduate from Telkom University with strong interests in Internet of Things (IoT), Front-End Development, and Deep Learning. Gained practical experience through laboratory assistantship and industry internships at PT LEN Industri and HUMIC Engineering. Adaptable, a collaborative team player, and ready to contribute professionally to the technology industry.

Work Experience

Project Management Officer, Internship **Jun 2025 - Sept 2025**
PT LEN Industri (Persero), Bandung

- Developed and managed project dashboards using Google Looker Studio for dynamic, real-time project monitoring with a focus on budget oversight.
- Utilized AppSheet and Looker to optimize project management processes during the internship period.
- Prepared various project documents required for management reporting.

Deep Learning Engineer, Internship **Apr 2026 - July 2026**
HUMIC (Human Centric) Engineering, Bandung

- Designed and implemented PyTorch Deep Learning architectures (Multi-Branch 1D/2D CNN & BiLSTM-MHA) for speech-based depression detection using the DAIC-WOZ dataset (189 participants) under strict participant-level 5-Fold Stratified Cross-Validation.
- Integrated self-supervised pre-trained representations (Wav2Vec 2.0 768-dim embeddings) with Mel-Spectrograms and 120-dim MFCCs into a multi-modal deep feature fusion head.
- Mitigated severe class imbalance (2.3:1 ratio) through custom AdamW optimization, ReduceLROnPlateau scheduling, 0.4 dropout regularization, and weighted Cross-Entropy Loss penalties.
- Engineered Explainable AI (XAI) attribution pipelines leveraging SHAP GradientExplainer directly on CNN fusion layers to interpret spectral-temporal audio feature attributions for clinical transparency.
- Co-authored an IEEE-formatted research paper: "A Comparative Study of Audio Feature Representations and Fusion Methods Across Machine Learning and Deep Learning for Depression Detection".

Project

ESP32 Based Smart Assistive Device for the Visually Impaired **Mar 2025 - Jun 2025**

- Designed a navigation aid for the visually impaired using an ESP32 microcontroller to enhance user safety and mobility.
- Integrated ultrasonic sensors for obstacle detection, a tilt sensor for balance monitoring, and a buzzer alarm system.
- Developed device logic using C++ and integrated it with the Blynk.io platform for monitoring functionality.

CryptoMania Website Development **Mar 2025 - Jun 2025**

- Built a cryptocurrency information platform providing real-time data on coin names, market values, descriptions, and price history.
- Led the front-end development using React.js, JavaScript, and CSS.
- Designed and implemented a responsive user interface to present data for the top 50 coins informatively.

A Computer Vision-Based Human Monitoring and Counting System Using YOLOv8n and ByteTrack **Feb 2026 - Aug 2026**

- Developed an edge AI human detection and counting system deployed on a Raspberry Pi 5 without personal identity recognition for user privacy.
- Implemented YOLOv8n fine-tuned on a custom dataset and integrated ByteTrack with virtual line-crossing logic to prevent double-counting.
- Built an asynchronous dual-path distribution to Firebase Realtime Database and created the Detectra mobile app using Flutter for live monitoring.

Organizational Experience

Public Relations Division, PERMISI **2022 - 2023**

- Actively participated as a team member in the Education Roadshow project targeting various high schools in the Bekasi area.
- Carried out intensive coordination and direct socialization of the program to schools to build and maintain strategic cooperative relationships.
- Managed internal team communication to ensure that activity schedules and presentation materials were delivered effectively and on time.
- Achievements: Successfully reached and increased the enthusiasm of high school students in Bekasi in learning about Telkom University's education programs through measurable outreach activities

Public Relations Division, TELERIUM **2023 - 2024**

- Acting as a member who serves as the main communication bridge through intensive coordination between laboratories within the Faculty of Electrical Engineering.
- Ensuring the smooth flow of information between the organizers and laboratory assistants in order to minimize technical obstacles during the event.

Practicum Division, MBC LABORATORY **2025 - 2026**

- Responsible for teaching practical material directly to students in the laboratory to ensure in-depth technical understanding.
- Compiling and developing practical modules as the main technical guide in the student learning process.
- Installing and configuring the necessary software to support the smooth running of practical activities.
- Managing the physical maintenance of the laboratory, including workbench layout and cable management, to ensure that the practical area remains safe and conducive.

Education

Telkom University **2022 - 2026**
S1 Telecommunications Engineering

Soft Skill

- Time Management
- Discipline
- Adaptability
- Teamwork

Hard Skill

- IoT Development: Arduino, ESP32
- Sensor Integration: Ultrasonic, Buzzer, Tilt Sensor
- Programming Languages (basic): C/C++, Python, HTML, CSS, JavaScript, MySQL
- Networking: Basic network configuration using Cisco Packet Tracer
- Microsoft Excel (basic)
- Google Looker Studio
- Computer Vision
- Deep Learning & AI Artificial Intelligence
- Explanaible AI
- DevOps : (Git, Github, Docker, Portainer, Jenkins, Sonarqube, Prometheus, Grafana)

Licenses & Certifications

Learn the Basics of AI, Dicoding Indonesia **Nov 2024 - Nov 2027**

Getting Started with Python Programming, Dicoding Indonesia **Nov 2024 - Nov 2027**

Learn the Basics of Data Science, Dicoding Indonesia **Oct 2024 - Oct 2027**

Learn the Basics of Structured Query Language (SQL), Dicoding Indonesia **Oct 2024 - Oct 2027**

Deep Learning, DeepLearning.AI **Aug 2026**

DevOps Training, ID-Network **Oct 2026**